Yousef Alsayed Abdelfattah

Cairo, Egypt | yousefabuelkhear@gmail.com | +20 115 382 6643 | linkedin.com/in/yousefalsayed-abdelfattah | github.com/yousefattaff

PROFESSIONAL SUMMARY

AI Engineer with a B.Sc. in Mechatronics, Robotics, and Automation Engineering. Experienced in building RAG pipelines, fine-tuning LLMs, and multi-agent systems using LangChain, LangGraph, Hugging Face, and modern agentic frameworks.

EXPERIENCE

AI Software Engineer *Nov 2024 – Jan 2025*
Outlier AI • Remote

- Evaluated and ranked LLM-generated responses across diverse domains to improve model accuracy and robustness.
- Identified failure modes and collaborated with AI trainers to enhance prompt-response alignment.

PLC Automation Engineer (Internship) *Sep 2023*
Siemens • Cairo, Egypt

- Designed and implemented industrial automation logic using Siemens TIA Portal for S7-1200 and S7-1500 PLCs with Ladder Logic programming.
- Debugged and troubleshot automation processes, gaining hands-on experience with real-world factory workflows and system integration.

CNC Programming & Operation (Internship) *Sep 2022*
DMG MORI • Cairo, Egypt

- Developed G-code programs for turning and milling operations; set up and operated CNC machines to precision engineering standards.

PROJECTS

Price Prediction & Autonomous Deal Agent
Built a price prediction pipeline from short product descriptions, fine-tuned an open-source LLM with LoRA/PEFT via Hugging Face, and engineered a multi-agent system (ReAct-style reasoning) that scans listings, detects pricing anomalies, and sends real-time deal alerts.

Medical RAG AI Agent
Architected an end-to-end RAG pipeline using LangChain, ChromaDB, and Hugging Face sentence-transformers, integrated Groq and Ollama (Llama 3.2) as interchangeable LLM backends, added agentic tool-use with function calling (web search, PubMed, OCR), and delivered a Streamlit web UI with text-to-speech output.

R2-Waiter
Built a PyQt5 desktop application as graduation project, integrating an animated robot face (HTML/CSS/JS) reacting live to ROS velocity commands with eye tracking and facial expressions, combined with a Qt order management interface supporting tray and table selection.

EDUCATION

Bachelor of Science – Mechatronics, Robotics, and Automation Engineering *Sep 2019 – Jun 2024*
Misr University for Science and Technology (MUST) • Giza, Egypt

CERTIFICATIONS

- LLM Engineering: Master AI & Large Language Models – Ed Donner / O'Reilly (2026)
- AI For Everyone – DeepLearning.AI / Coursera (2025)
- CS50's Introduction to Computer Science (2023) / CS50's Introduction to Artificial Intelligence (2025) – Harvard / edX
- Embedded Systems: AVR – Independent (2023) | EF SET English Certificate – C2 Proficient (71/100)

SKILLS

LLMs & AI: RAG Pipelines, AI Agents, Agentic Reasoning (ReAct, Tool Calling), Structured Outputs / JSON Mode, Prompt Engineering, Fine-tuning (LoRA/PEFT), Embeddings & Semantic Search

Frameworks: LangChain, LangGraph, CrewAI, AutoGen, LangSmith, FastAPI, Streamlit, Pydantic

Vector Databases: ChromaDB, Sentence-Transformers, Semantic Search, Embedding Models

Frontend: React, Tailwind CSS, Streamlit, HTML/CSS/JS

Languages: Python (primary), C

Tools & DevOps: Git, GitHub, Docker, uv, pip

LANGUAGES

ARABIC: Native **ENGLISH:** C2 Proficient